

Lab 3: Component Modelling & Architectural Pattern Selection

Architecture Selection

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

Architectural Choice

Layered Architecture organizes the system into logical layers, separating the user interface, business logic, payment processing, hardware interaction, and data storage. The system uses the Kiosk UI, Order Manager, Payment Service, Menu & Pricing Database, and Receipt Printer components.

Reason 1 – Separation of Concerns

Each component has a specific responsibility. The Kiosk UI handles touchscreen interactions, while the Order Manager processes orders. The Payment Service handles credit card payments, and the database stores menu and pricing information.

Reason 2 – Easy Maintenance

The modular structure makes the system easier to maintain and modify. For example, the receipt printer component can be replaced with compatible hardware without redesigning the entire application.

Security Advantage

The Payment Service can be separated from the user interface, limiting direct access to payment-processing functions. Payment data should be transmitted securely, and sensitive card details should not be stored unnecessarily.

Performance Benefit

Separating the user interface from order processing and data storage helps organize the workload. Menu and pricing information can be retrieved when needed, allowing the system to keep touchscreen interactions responsive.

